

ISM4641: Python for Business Analytics

CASE 01: The Cozy Coffee Corner Company

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Business Scenario: “The Cozy Coffee Corner”

Story and Background:

The Cozy Coffee Corner is a small, independent coffee shop that has been gaining popularity in the neighborhood. They pride themselves on their high-quality coffee, friendly service, and cozy atmosphere. However, the owner, Sarah, has noticed that while they have a steady stream of customers, she’s not sure which products are performing best, who her most valuable customers are, or how effective her recent marketing efforts have been.

Sarah has been diligently collecting data at the point of sale, but it’s all stored in a simple spreadsheet. She knows she needs to analyze this data to make informed business decisions, but she’s overwhelmed by the prospect of doing it manually. **Luckily, she’s heard that you’re taking a Python for Business Analytics course and has come to you for help! Data Available to Students:**

Sarah has provided you with the following data:

```
1  # Data for The Cozy Coffee Corner
2
3  # Prices of items on the menu
4  prices = {
5      "Espresso": 2.50,
6      "Latte": 3.50,
7      "Croissant": 2.00
8  }
9
10 # Orders for one day (list of lists, where each inner list represents an order)
11 orders = [
12     ["Espresso", "Latte", "Croissant"],
13     ["Espresso", "Espresso", "Croissant"],
```

```

14     ["Latte", "Latte"],
15     ["Espresso", "Croissant", "Croissant"]
16 ]
17
18 # Customer information (list of dictionaries, each representing a customer)
19 customers = [
20     {"name": "Alice", "member": True},
21     {"name": "Bob", "member": False},
22     {"name": "Charlie", "member": True}
23 ]

```

The Business Problem:

Sarah wants you to help her answer these questions using Python:

1. What is the total sales for the day?
2. What is the average order value for the day?
3. How many customers are members of the loyalty program?

Applying Week 1 and 2 Concepts:

Here's how you can use the concepts from the first two weeks to solve this problem:

1. Data Structures:

- **prices (Dictionary):** Stores the price of each item.
- **orders (List of Lists):** Represents the orders for the day.
- **customers (List of Dictionaries):** Represents the customers.

2. Control Flow:

- **for loops:**
 - Iterate through the **orders** list.
 - Iterate through the items within each order.
 - Iterate through the **customers** list.
- **if statements:** (Optional) Check if an item exists in **prices**.
- **if statements** Check if a customer is a member.

3. Variables and Data Types:

- **str:** Store item names, customer names.
- **float:** Store prices and calculate totals.

- **bool:** Store customer membership status.
- **int:** Store the number of members

4. Operators:

- **Arithmetic operators (+):** Calculate order totals and the total sales.
- **Comparison operators (==):** Check for membership status.

5. Basic Calculations:

- **Total sales:** Sum the totals of all orders.
- **Average order value:** Divide the total sales by the number of orders.
- **Number of members:** Count the number of customers who are members.

Sample Code (Complete Solution):

```

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14     ["Latte", "Latte"],
15     ["Espresso", "Croissant", "Croissant"]
16 ]
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18 # Customer information (list of dictionaries, each representing a customer)
19 customers = [
20     {"name": "Alice", "member": True},
21     {"name": "Bob", "member": False},
22     {"name": "Charlie", "member": True}
23 ]
24
25 # 1. Calculate total sales for the day
26 total_sales = 0

```

```

27 for order in orders:
28     order_total = 0
29     for item in order:
30         order_total += prices[item] # Look up the price in the `prices` dictionary
31     total_sales += order_total
32
33 print(f"1. Total sales for the day: ${total_sales:.2f}")
34
35 # 2. Calculate the average order value
36 average_order_value = total_sales / len(orders)
37 print(f"2. Average order value: ${average_order_value:.2f}")
38
39 # 3. Count the number of members
40 member_count = 0
41 for customer in customers:
42     if customer["member"]:
43         member_count += 1
44
45 print(f"3. Number of members: {member_count}")

```

Expected Output:

```

1. Total sales for the day: $28.50
2. Average order value: $7.12
3. Number of members: 2

```

Congratulations:

By applying these concepts, you have written a Python program that analyzes the provided data, answers Sarah's questions, and provides her with valuable insights to improve her business. This exercise is meant to help reinforce your understanding of fundamental Python concepts and demonstrate how they can be applied to solve real-world business problems.